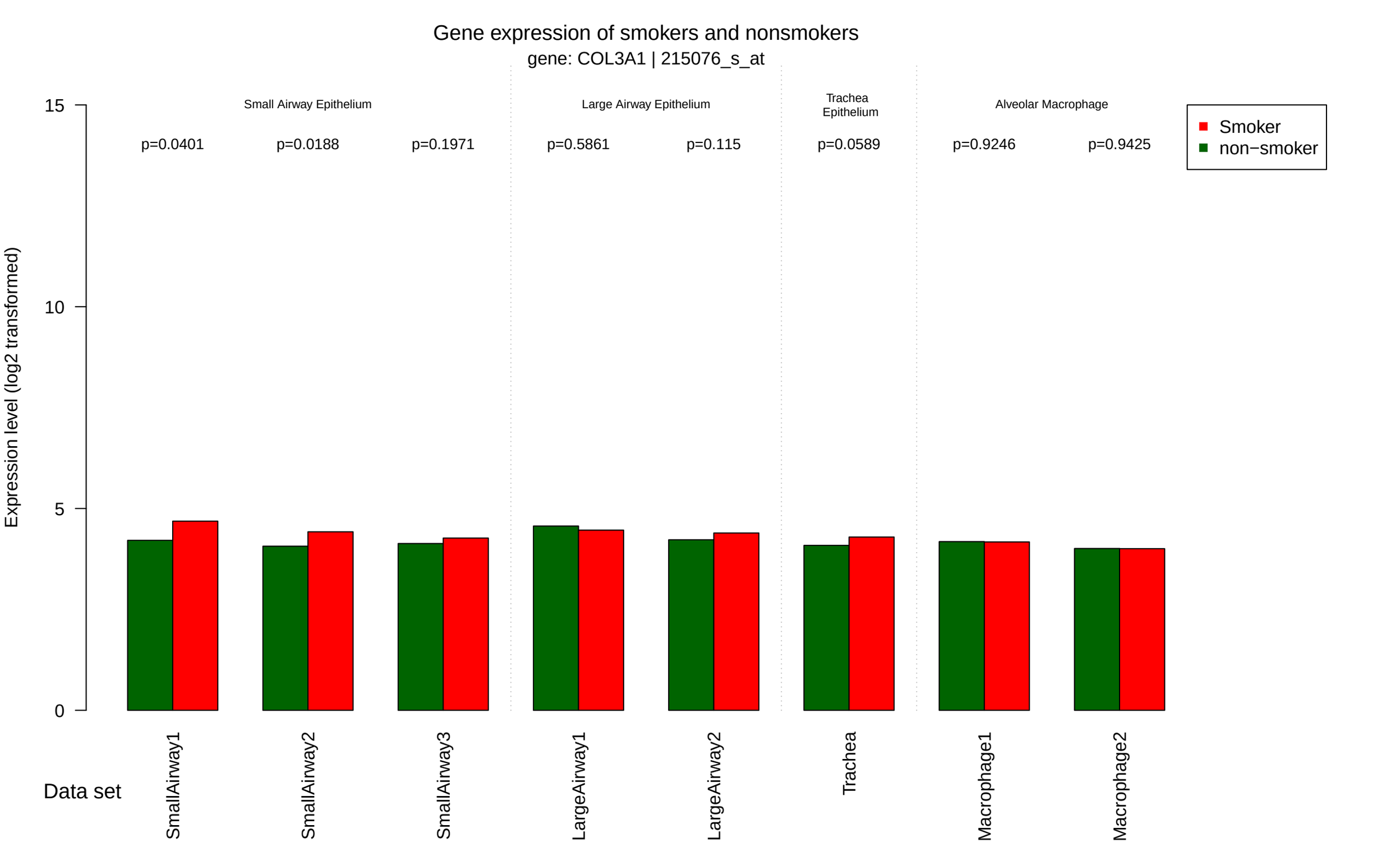
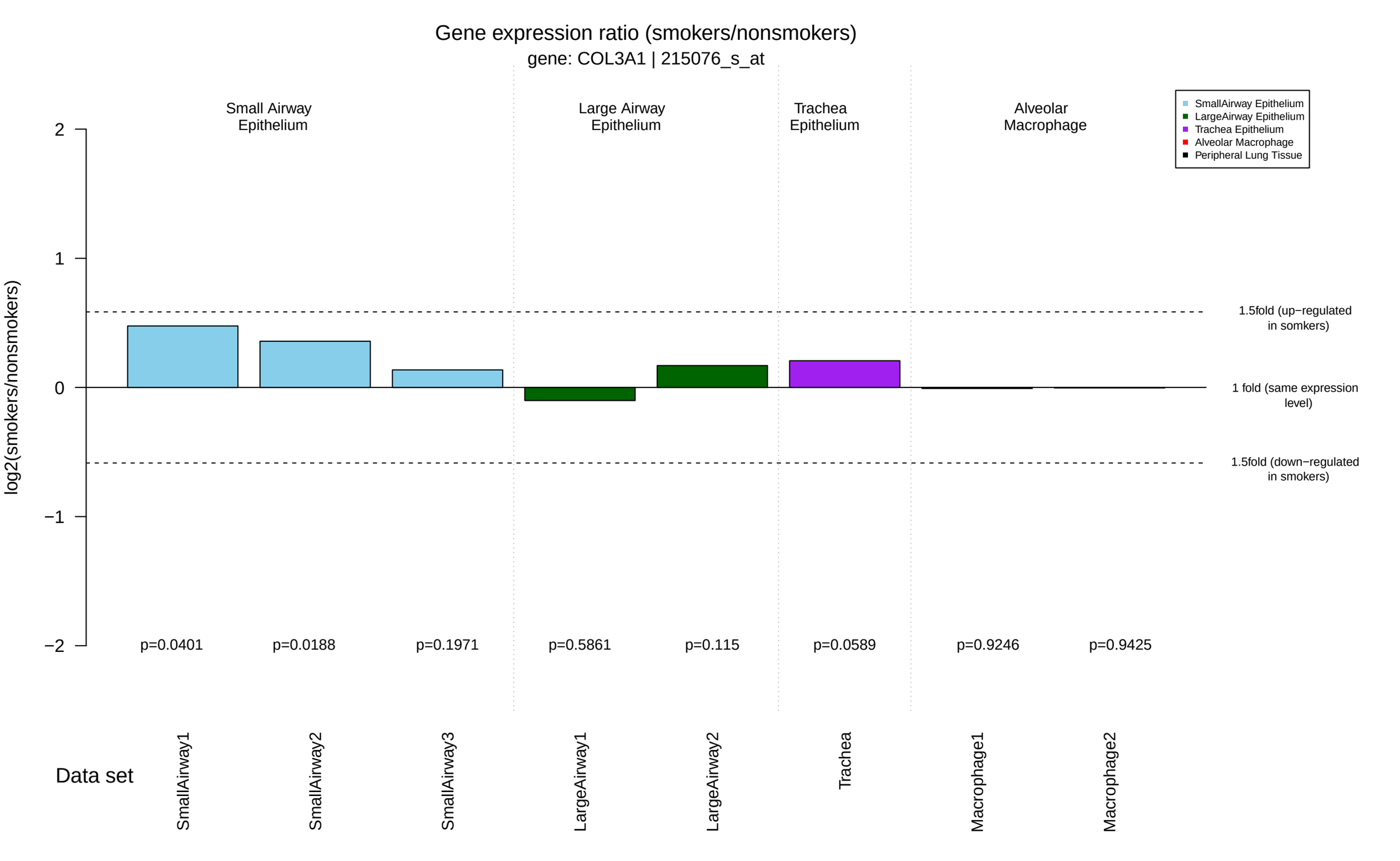
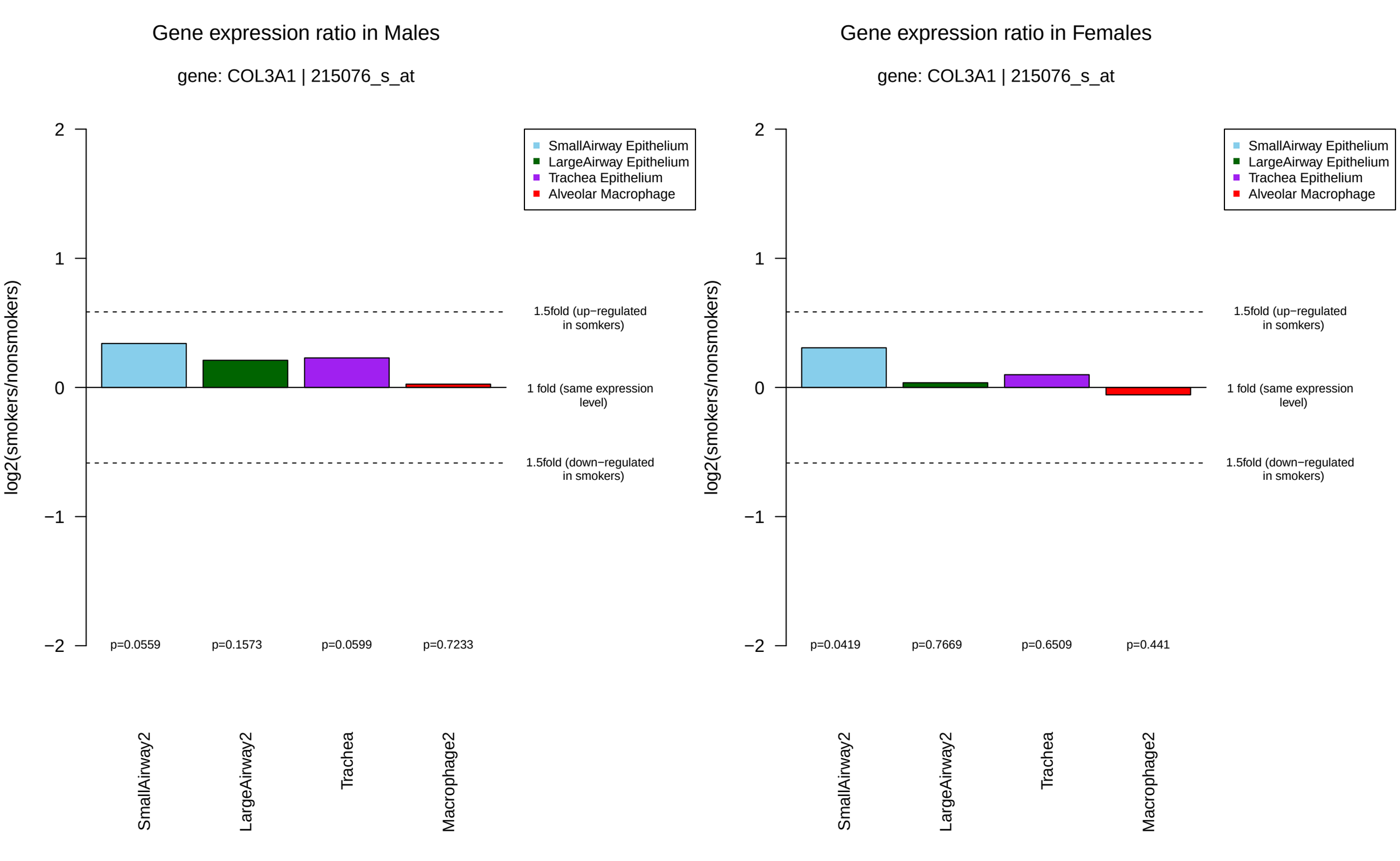
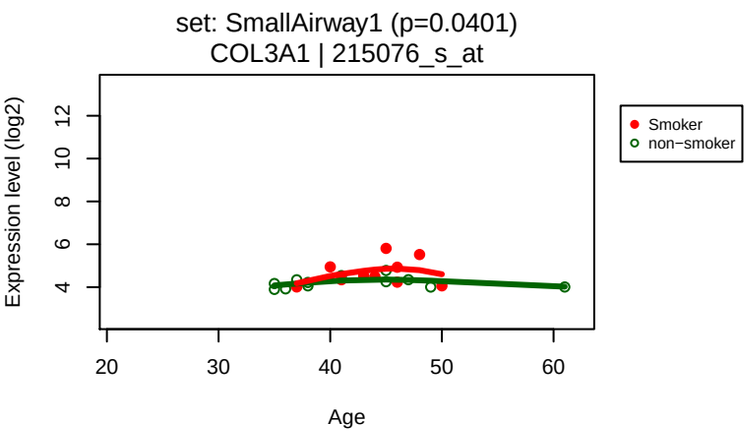
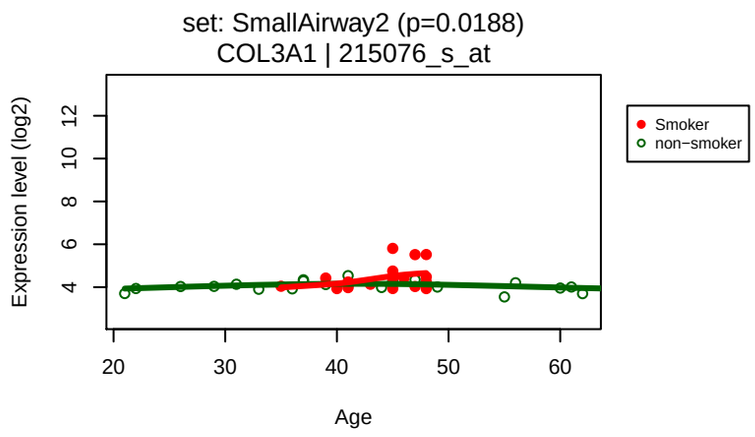
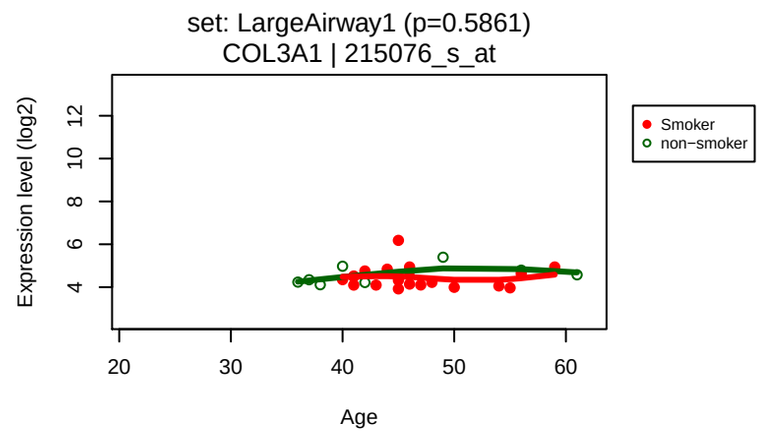

Gene expression of smokers and nonsmokers
gene: COL3A1 | 215076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0401
p=0.0188
p=0.1971
p=0.5861
p=0.115
p=0.0589
p=0.9246
p=0.9425
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

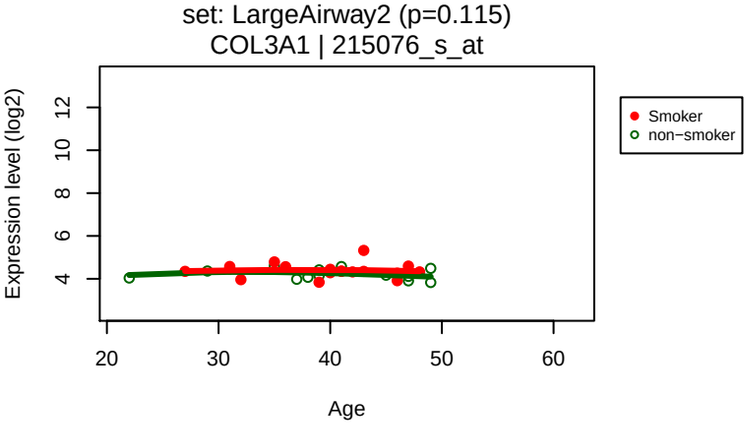

Gene expression ratio (smokers/nonsmokers)
gene: COL3A1 | 215076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0401
p=0.0188
p=0.1971
p=0.5861
p=0.115
p=0.0589
p=0.9246
p=0.9425
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

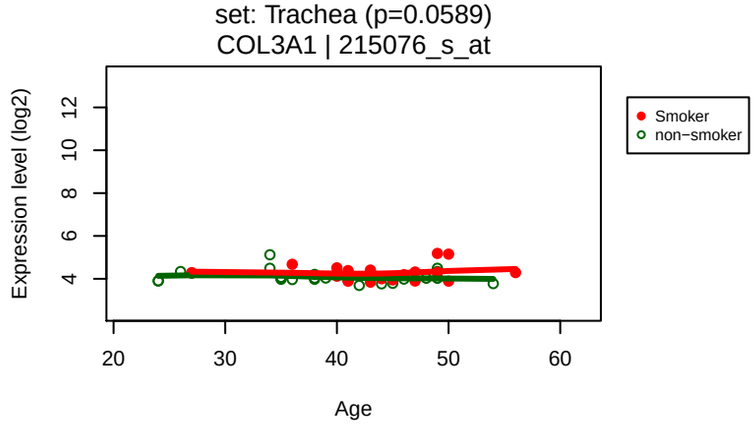

Gene expression ratio in Males
gene: COL3A1 | 215076_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0559
p=0.1573
p=0.0599
p=0.7233
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL3A1 | 215076_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0419
p=0.7669
p=0.6509
p=0.441
SmallAirway2
LargeAirway2
Trachea
Macrophage2

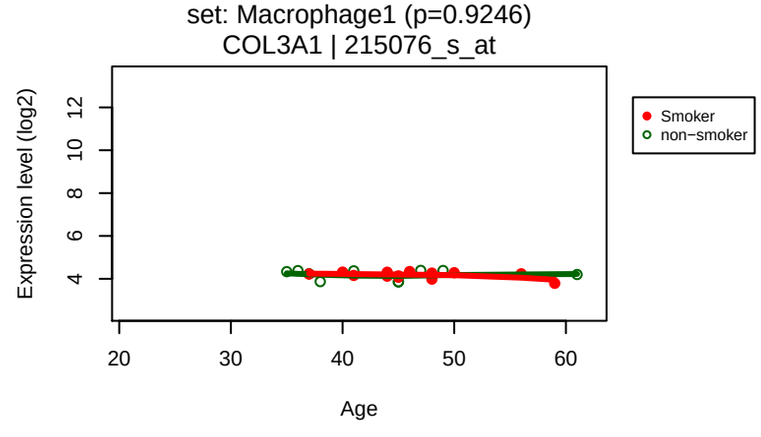

set: SmallAirway1 (p=0.0401)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker

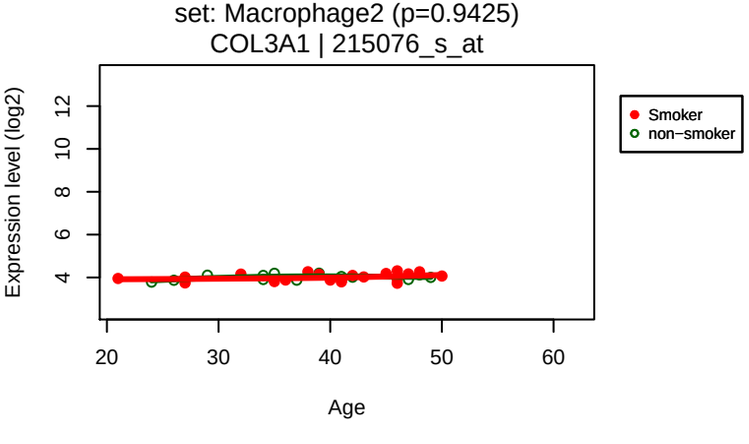

set: SmallAirway2 (p=0.0188)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker

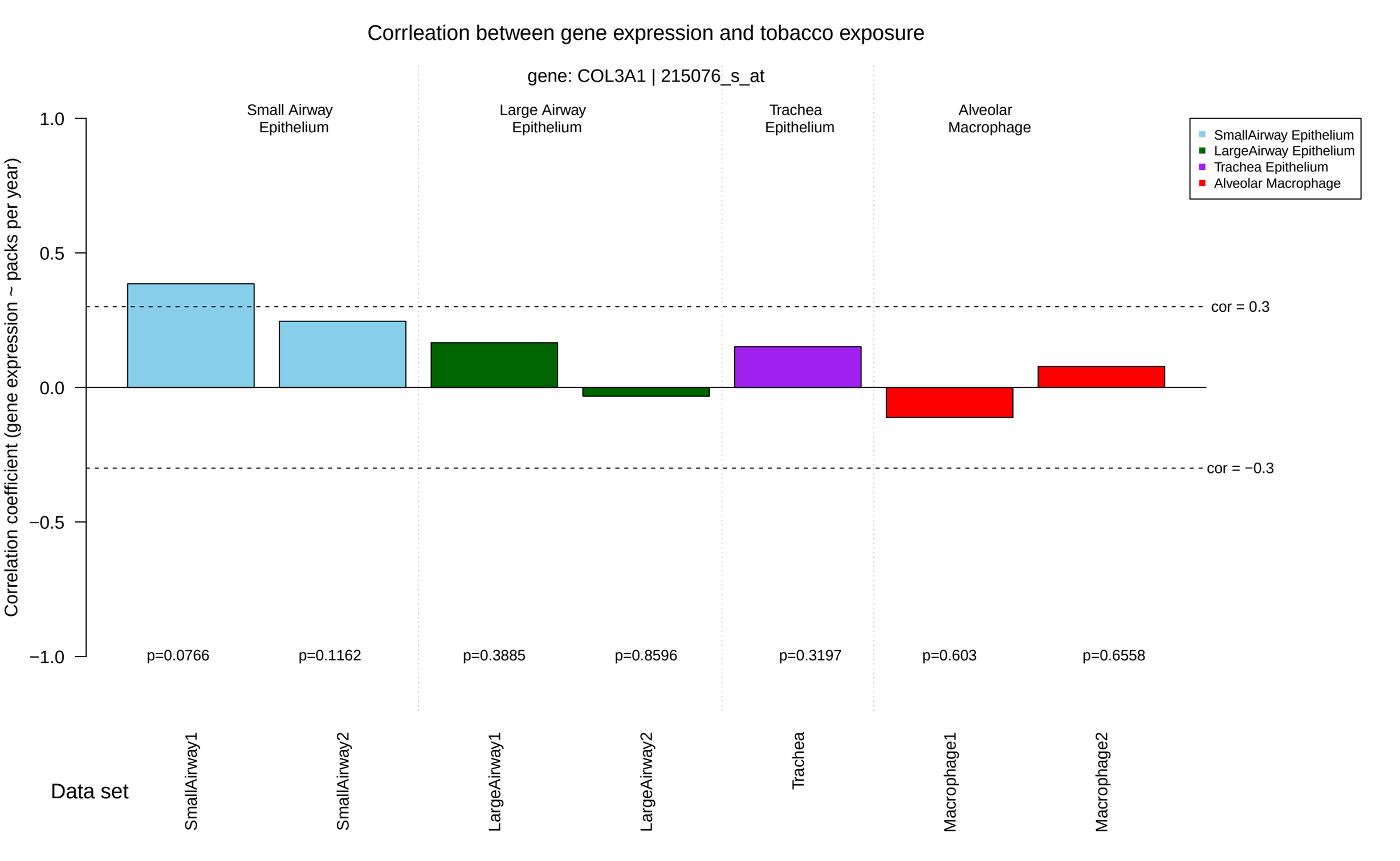

set: LargeAirway1 (p=0.5861)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.115)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0589)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9246)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9425)
COL3A1 | 215076_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL3A1 | 215076_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0766
p=0.1162
p=0.3885
p=0.8596
p=0.3197
p=0.603
p=0.6558
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2